

Survival, Anatomic, and Functional Long-term Results in Choroidal and Ciliary Body Melanoma After Ruthenium Brachytherapy (15 Years' Experience With Beta-rays)

F. ROUBEROL, MD, P. ROY, MD, PhD, L. KODJIKIAN, MD, J. P. GÉRARD, MD,
B. JEAN-LOUIS, MD, AND J. D. GRANGE, MD

• **PURPOSE:** To analyze survival, radiation-related complications, local recurrence, enucleation, and visual acuity after ruthenium-106 irradiation of malignant uveal melanoma.

• **DESIGN:** Retrospective study.

• **METHODS:** A total of 213 patients treated with ruthenium-106 brachytherapy between 1983 and 1995. End point rates were estimated by life tables, and prognostic factors by Cox proportional hazards regression. Main outcome measures were survival rate, radiation-related maculopathy, radiation-related vascular occlusion, local tumor recurrence, enucleation, and visual acuity (<20/100 at 3 years).

• **RESULTS:** At 5 and 10 years, survival was 82% (standard error [SE] 2.7%) and 72% (SE 3.4%), local recurrence 21.7% (SE 3.0%) and 24.3% (SE 3.2%), and enucleation 18.0% (SE 2.7%) and 19.2% (SE 2.8%), respectively. Sixty-one patients showed maculopathy (29%), 36 retinal vascular occlusion (17%), in 33 local recurrence (16%), and 38 enucleation (18%). Age and large tumor diameter were independently associated with survival ($P < .0001$ and $P < .0075$, respectively). Age below 40 and melanoma located posteriorly were significant risk factors for maculopathy ($P < .0085$ and $P < .0004$, respectively) and vascular occlusion ($P < .0415$ and $P < .0114$, respectively). Diameter and Bruch membrane rupture were significant predictors ($P <$

.0032 and $P < .0390$, respectively) of local recurrence. Visual acuity <20/100 was observed in only 26 of 97 (27%) cases of anterior but 34 of 42 (81%) of posterior tumor ($P < .001$).

• **CONCLUSION:** Although percentage tumor recurrence was high, survival was comparable to series using other treatments. Radiation-related complication rates were acceptable, especially for anterior tumors. Ruthenium therapy can, therefore, be recommended for small and medium-sized tumors with anterior location. (Am J Ophthalmol 2004;137:893–900. © 2004 by Elsevier Inc. All rights reserved.)

EPISELERAL PLAQUE RADIOTHERAPY HAS BEEN USED IN the treatment of posterior uveal melanoma for over 30 years and is an effective method for controlling this tumor locally.^{1–3} Most tumors respond to this treatment with a gradual decrease in size over a period of months to years.⁴ Ruthenium plaque radiotherapy for posterior uveal melanoma was introduced in the 1960s and subsequently popularized by Lommatzsch and associates (Lommatzsch & Vollmar 1966; Lommatzsch 1986) to avoid the side effects of cobalt-60. Brachytherapy with 106-ruthenium (106Ru/106Rh) has since been adopted in many ocular oncology centers.

The survival rate, the occurrence of complications, and possible prognostic factors for patients treated for uveal melanomas by ruthenium-106 brachytherapy have been addressed in several previous studies. We analyzed the survival, complications, rates of recurrence, and enucleation and functional results in successive patients initially treated by ruthenium plaque therapy during a 12-year period in one hospital. The risk factors associated with survival, maculopathy, vascular occlusion, and recurrence were analyzed by multivariate analysis.

Biosketches and/or additional material at www.ajo.com.

Accepted for publication Dec 8, 2003.

From the University Eye Clinic, Hôpital de la Croix Rousse, Lyon, France (F.R., L.K., B.J.-L., J.D.G.); and Departments of Biostatistics (P.R.), and Radiation Oncology (J.P.G.), Centre Hospitalo-Universitaire Lyon-Sud, Cedex, France.

Inquiries to Frédéric Roubérol, MD, University Eye Clinic, Hôpital de la Croix Rousse, 103 Grande rue de la Croix-Rousse, 69004 Lyon, France; e-mail: fredrouberol@yahoo.fr